



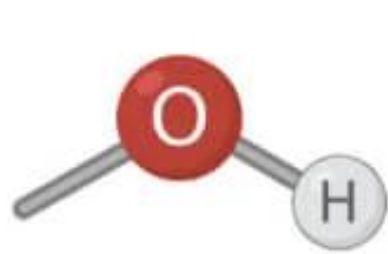
Lipids

Devanji Karunaratne

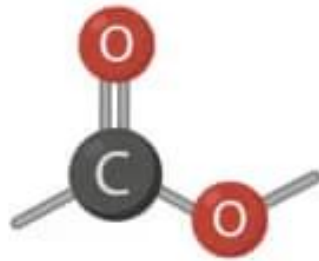
Intended Learning Outcomes

1. Explain the fundamental chemical properties defining lipids,
2. Classify lipids into structural categories and summarize their diverse functional footprints in cellular systems.
3. Analyze how hydrocarbon chain length and the degree of cis/trans saturation influence the physical melting points and molecular packing of fatty acids.
4. Distinguish between critical chemical reactions involving lipids.

Lipids- Definition, Structure and Functions, Fatty acids



Alcohol



Ester



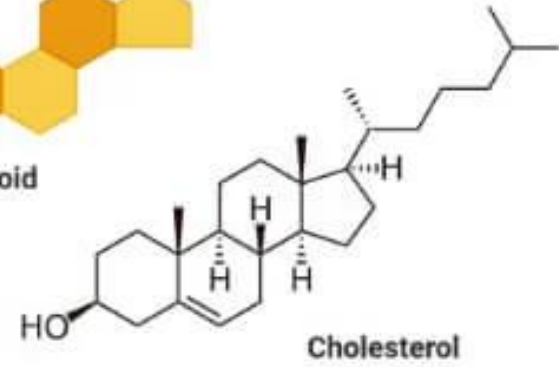
Saturated fatty acids



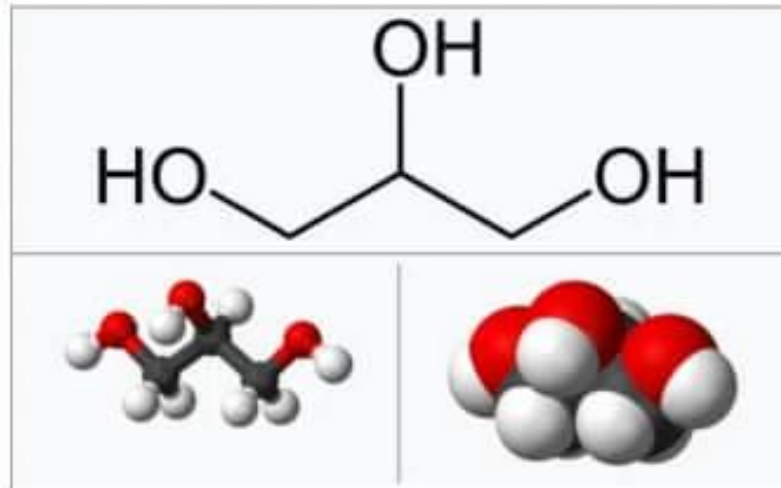
Unsaturated fatty acids



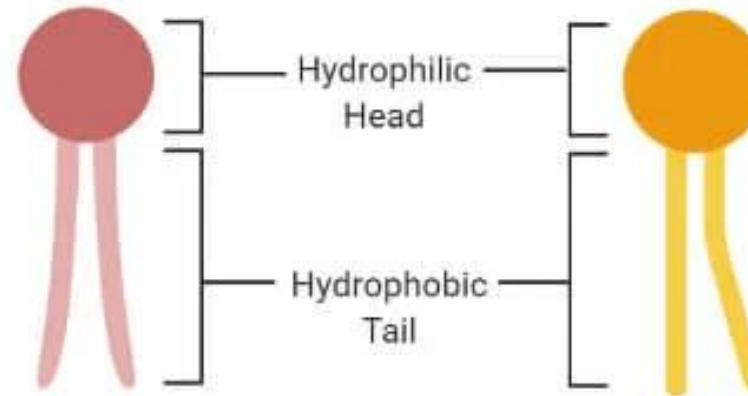
Steroid



Cholesterol



Glycerol



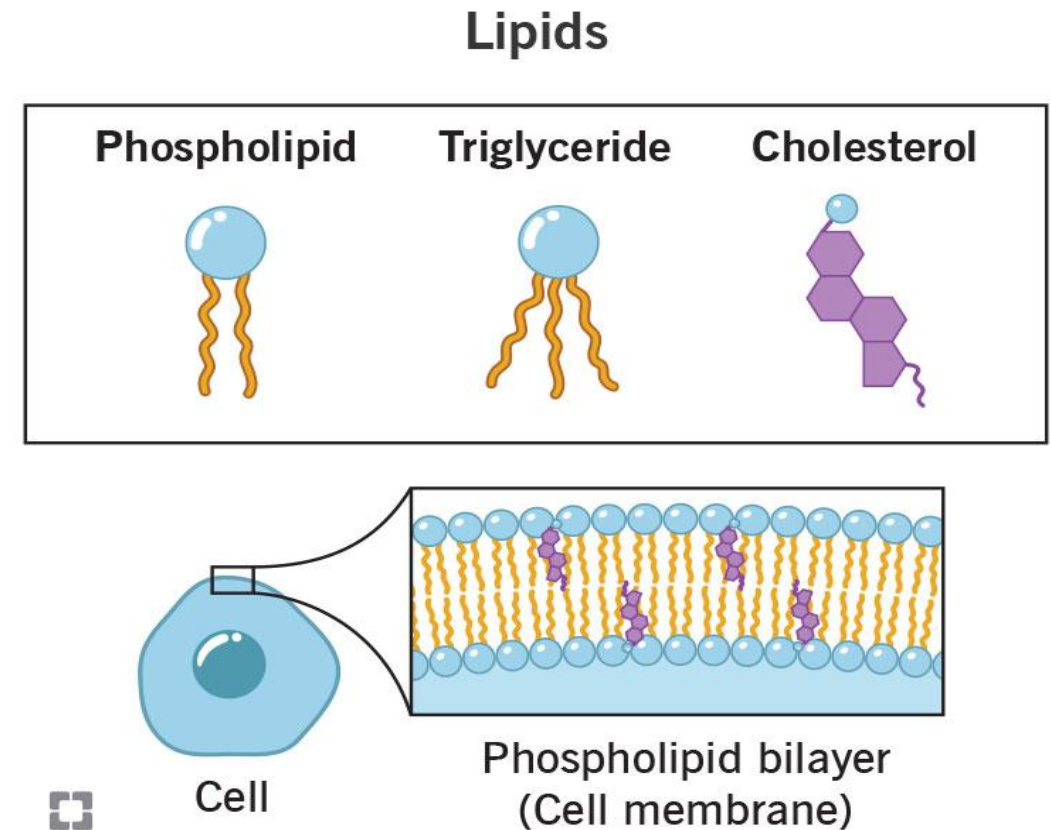
Phospholipids



Triglycerides

Introduction to Lipids

- Lipids are not defined by a common monomeric subunit.
- Instead, they are defined by a shared physical property: their **hydrophobicity**.



Key Defining Characteristics

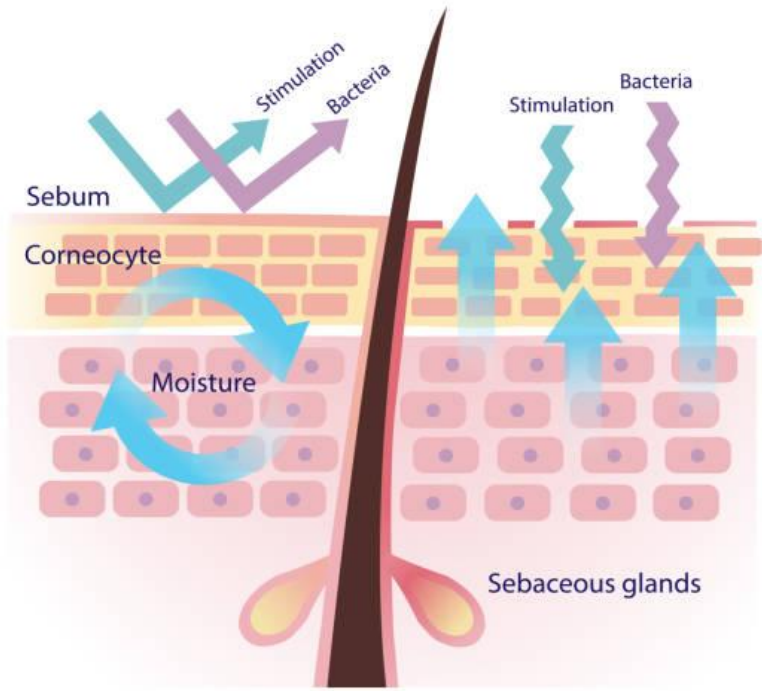
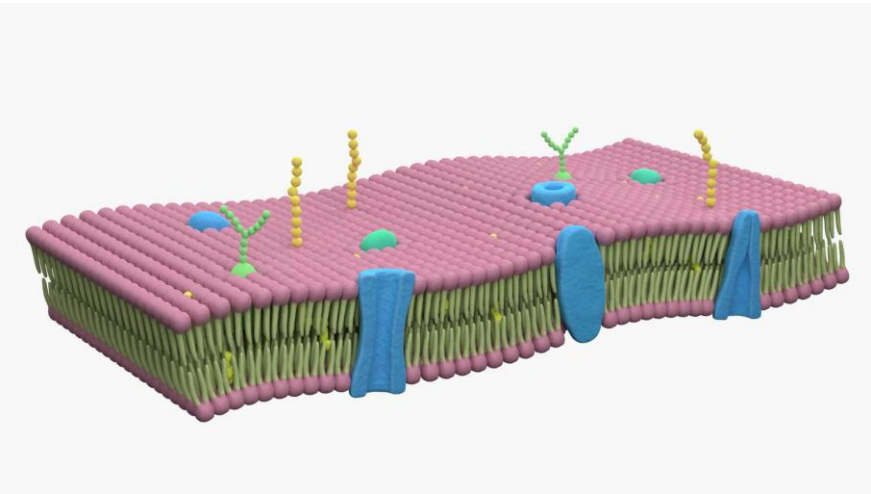
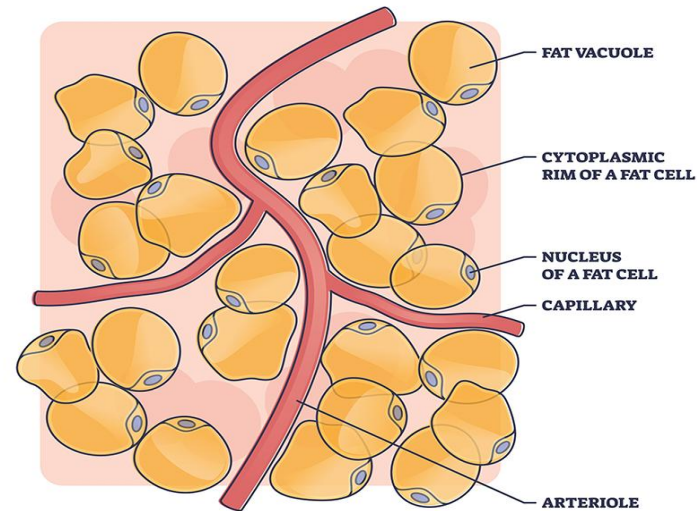
- **Insolubility in Water:** Lipids are largely nonpolar and therefore insoluble in polar solvents like water.
- **Solubility in Nonpolar Solvents:** They are soluble in organic solvents such as ether, chloroform, or benzene.



Crucial Functions of Lipids in Biological Systems

- **Energy Storage:** The most efficient form of long-term energy storage in animals (e.g., fats in adipose tissue).
- **Structural Components:** The fundamental building blocks of all biological membranes (the lipid bilayer).
- **Signaling Molecules:** Hormones and intracellular messengers (e.g., steroid hormones, prostaglandins).
- **Protective Coatings:** Provide a waterproof, protective layer for organisms (e.g., waxes on leaves, skin).
- **Vitamins and Pigments:** Some lipids act as essential vitamins (A, D, E, K) or pigments.

Adipose Tissue Body Fat



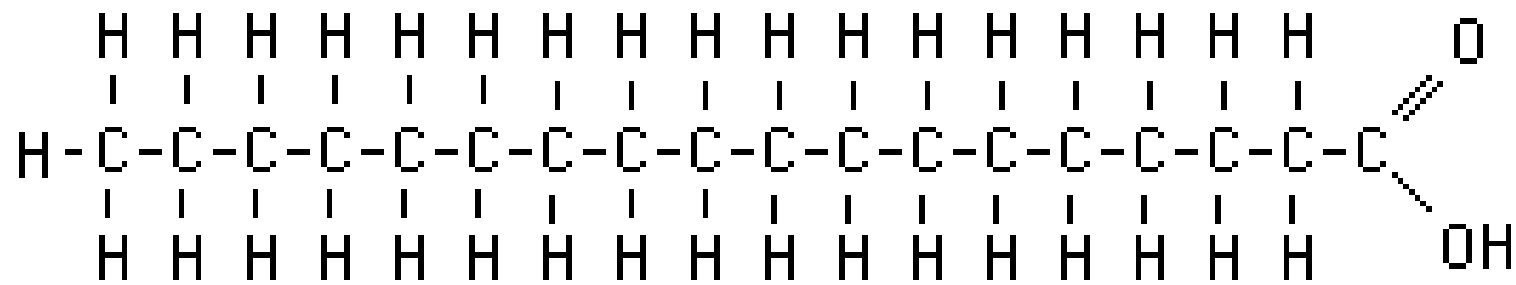
Healthy skin / Damaged skin

Classification of Lipids

A. Fatty Acids: The Fundamental Building Blocks

- Many lipids are derived from or contain **fatty acids**.
- A fatty acid is a carboxylic acid with a long hydrocarbon chain.
- It has two ends: a hydrophilic (polar) **carboxyl head** and a hydrophobic (nonpolar) **hydrocarbon tail**.

Structure: $\text{CH}_3-(\text{CH}_2)_n-\text{COOH}$



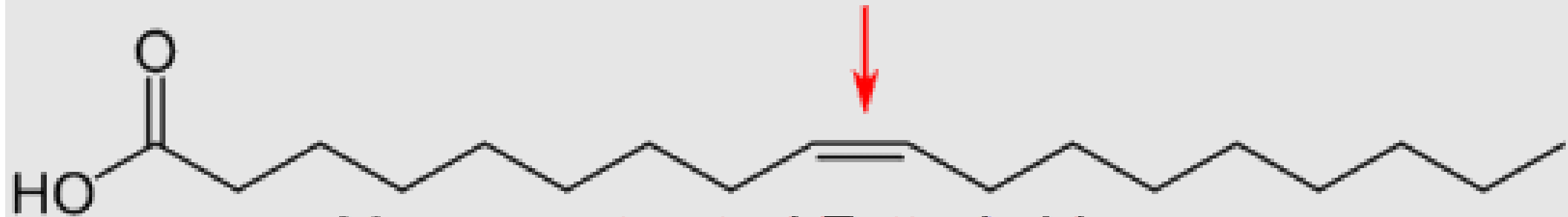
long hydrocarbon chain

carboxylic
acid group

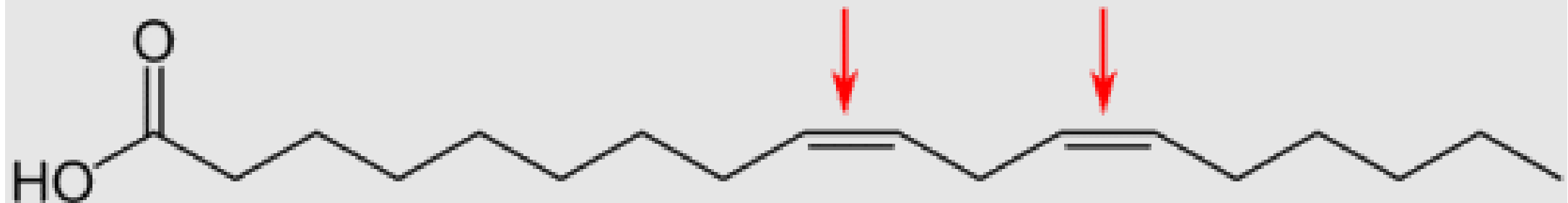
Essential features of a fatty acid



Palmitic Acid - Saturated Fatty Acid



Oleic Acid - Monounsaturated Fatty Acid



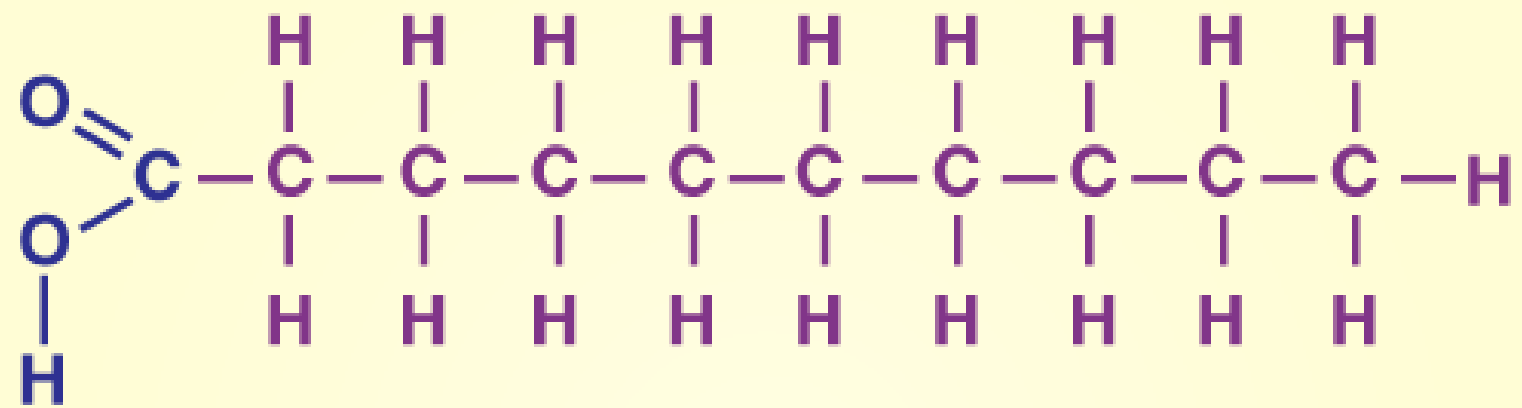
Linoleic Acid - Polyunsaturated Fatty Acid

A. Fatty Acids

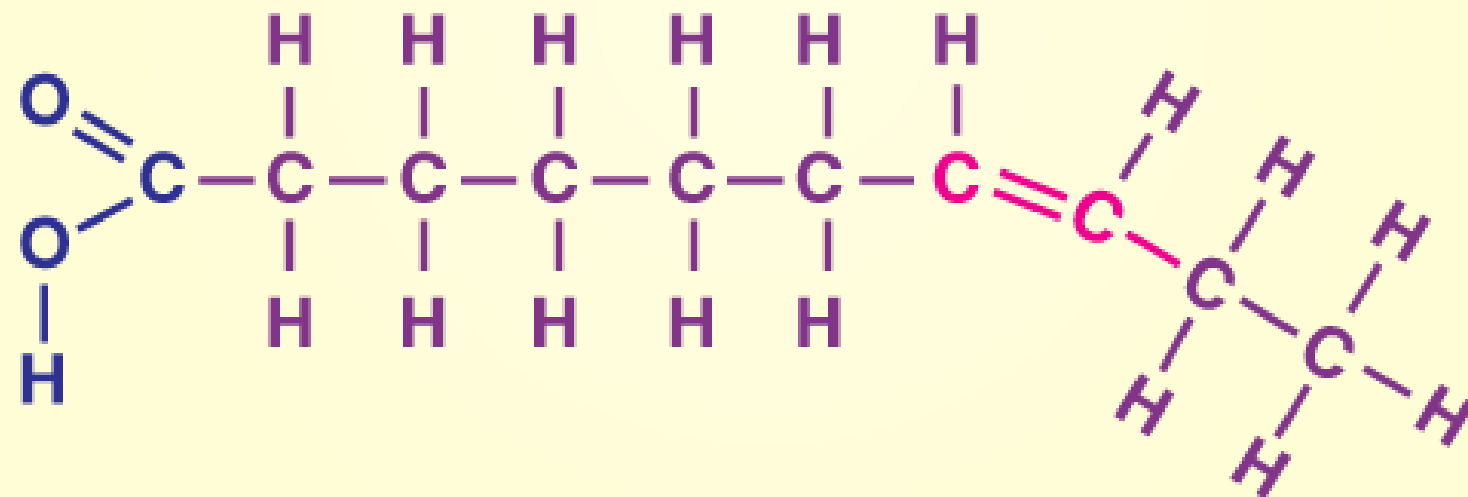
Saturation: This refers to the number of double bonds in the hydrocarbon chain.

- **Saturated Fatty Acids:** Contain **no carbon-carbon double bonds**. The chains are straight and flexible, allowing them to pack tightly together. They are typically **solid at room temperature**.
Example: Palmitic acid (C16), Stearic acid (C18).
- **Unsaturated Fatty Acids:** Contain one or more **carbon-carbon double bonds**.
 - **Monounsaturated:** One double bond (e.g., Oleic acid, found in olive oil).
 - **Polyunsaturated:** Two or more double bonds (e.g., Linoleic acid, found in plant oils).

(a) Saturated



(b) Unsaturated

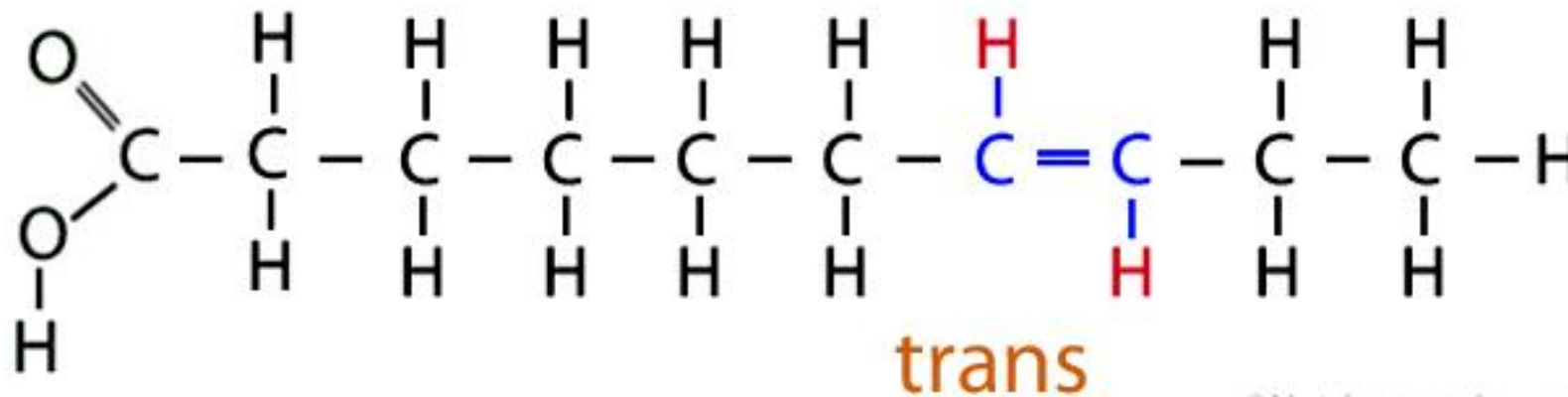
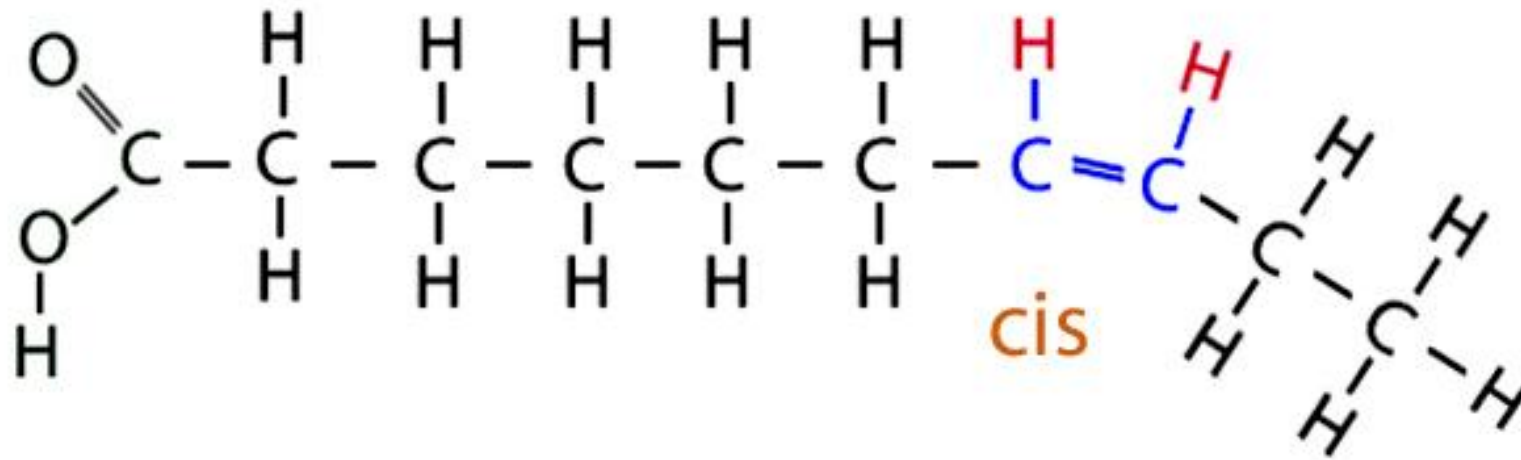


A. Fatty Acids

Geometric Isomerism (Cis vs. Trans)

- **Cis configuration:** The hydrogen atoms on the double-bonded carbons are on the **same side**. This creates a rigid "**kink**" or bend in the fatty acid chain. This kink prevents tight packing and lowers the melting point. Most naturally occurring unsaturated fatty acids are in the cis configuration.
- **Trans configuration:** The hydrogen atoms are on **opposite sides**. This allows the chain to remain relatively straight, similar to a saturated fatty acid. Trans fats are often produced artificially during the hydrogenation of vegetable oils and are linked to cardiovascular disease.

Cis- and Trans-Fatty Acids



A. Fatty Acids

Nomenclature:

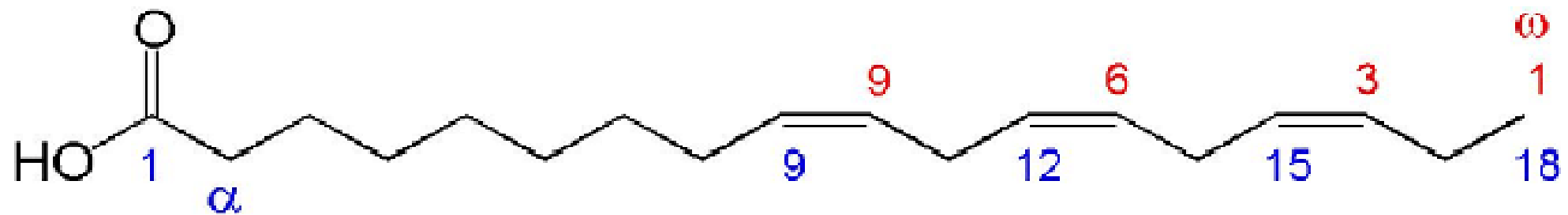
- **Δ (Delta) System:** Numbering starts from the carboxyl carbon. A fatty acid with a double bond at carbon 9 is denoted as $\Delta 9$.
- **ω (Omega) System:** Numbering starts from the methyl carbon (the last carbon in the chain). An $\omega-3$ fatty acid has a double bond at the third carbon from the methyl end. This system is useful for nutrition, as certain $\omega-3$ and $\omega-6$ fatty acids are essential for humans.

Number of carbons

Position of the first double bond from the methyl end

18:3 n-3

Number of double bonds



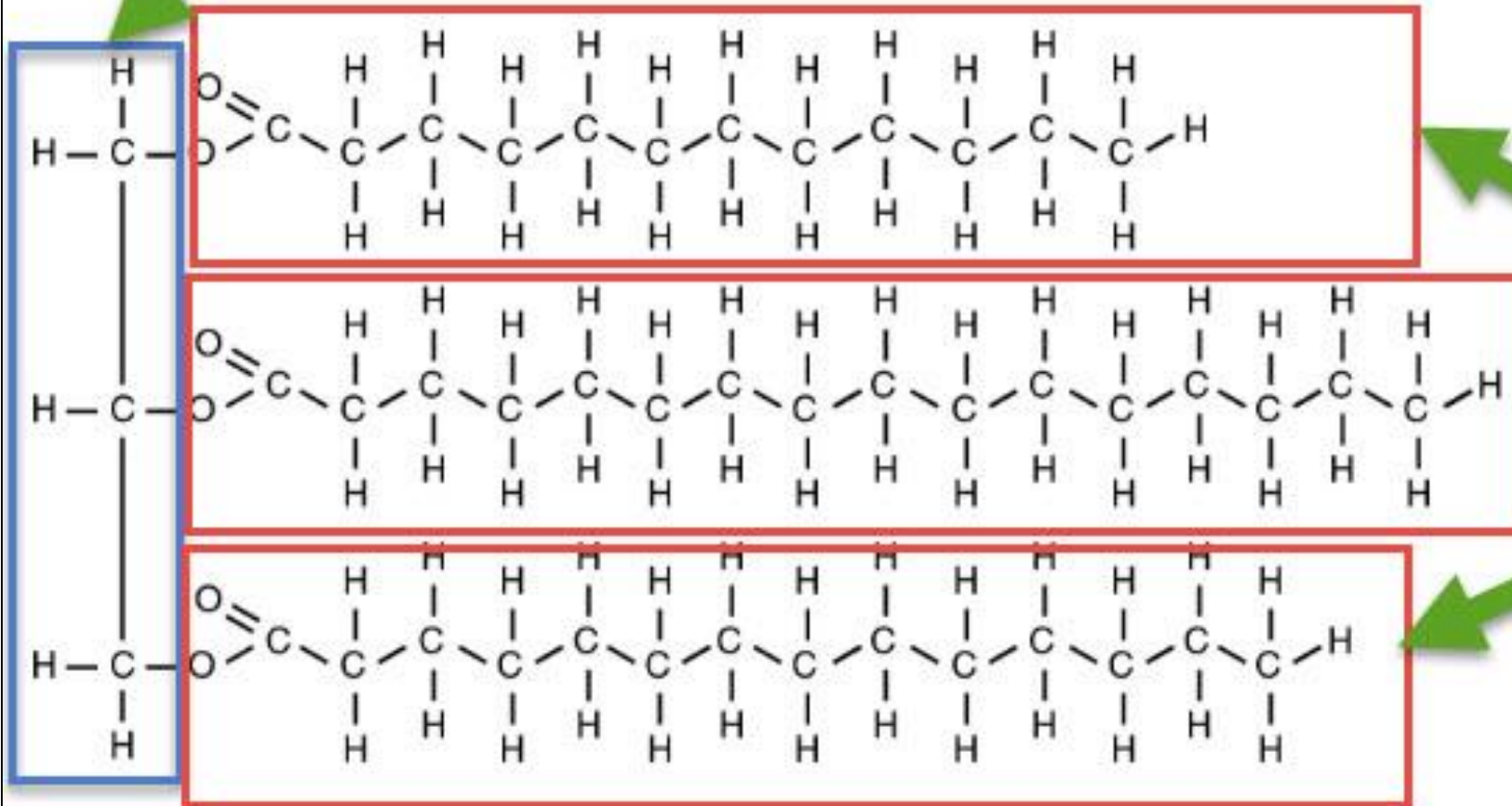
B. Simple Lipids (Fatty Acyl Esters)

These lipids are esters of fatty acids with various alcohols.

1. Triacylglycerols (Triglycerides):

- **Structure:** Composed of a **glycerol** backbone (a three-carbon alcohol) to which **three fatty acids** are attached via **ester bonds**. The fatty acids can be the same or different.
- **Function:** This is the primary form of long-term energy storage in animals and plants. They are stored in adipose tissue.
- **Properties:** Triacylglycerols are completely nonpolar, hydrophobic, and neutral (no charge), making them an ideal form of stored energy. They are very compact and highly reduced, yielding more than twice the energy per gram compared to carbohydrates.

Glycerol

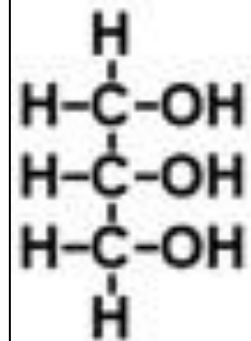
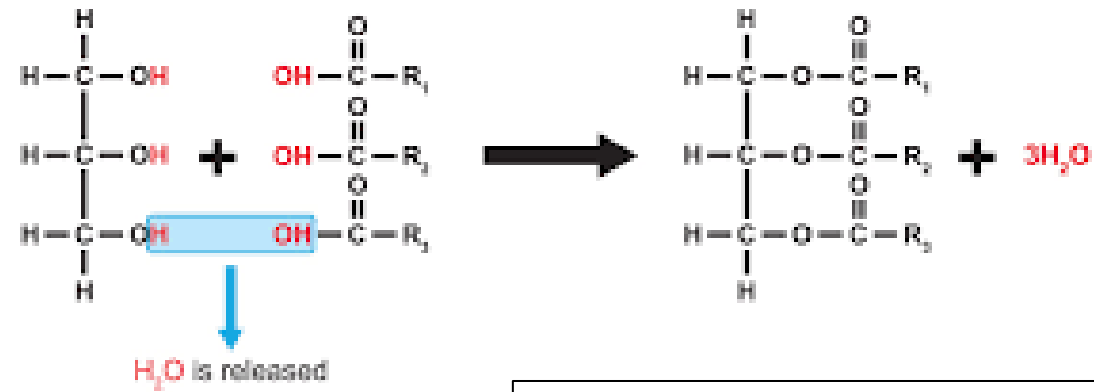


Fatty
Acids

Glycerol

3 Fatty Acids

Triglycerides



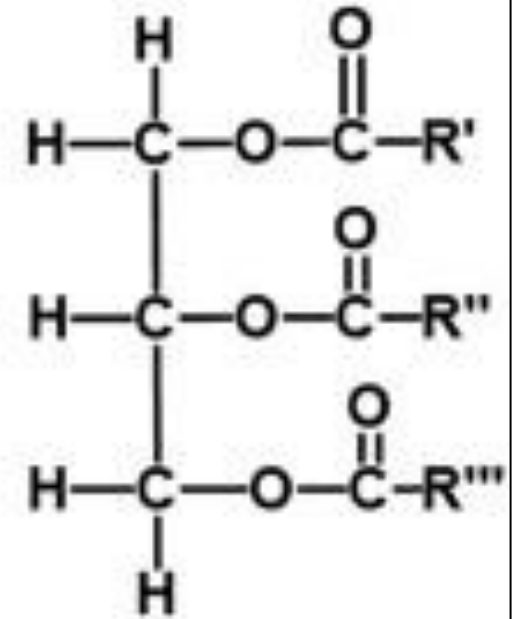
Glycerol

$\text{R}'\text{COOH}$

$\text{R}''\text{COOH}$

$\text{R}'''\text{COOH}$

Fatty acids



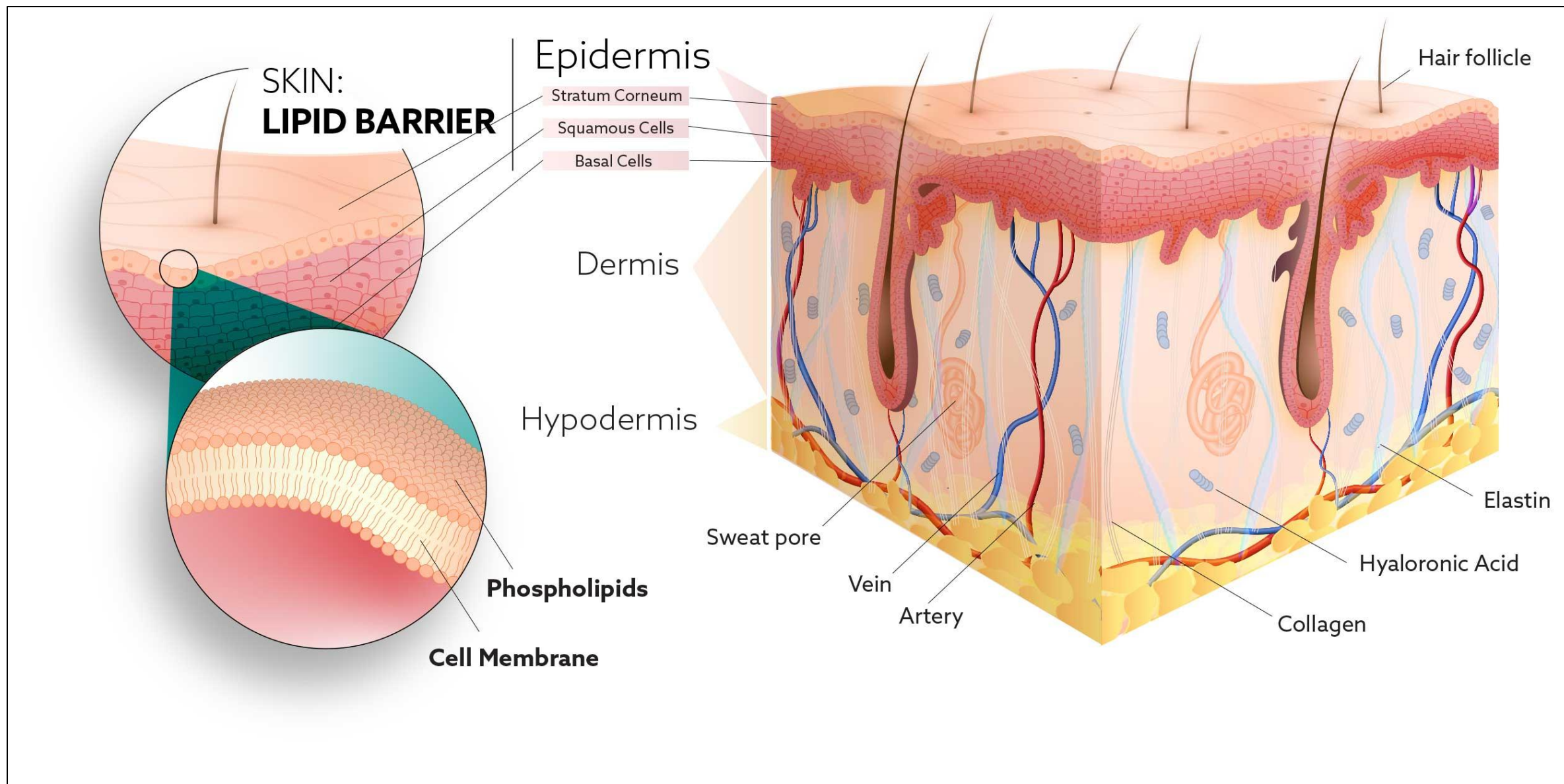
Triglyceride

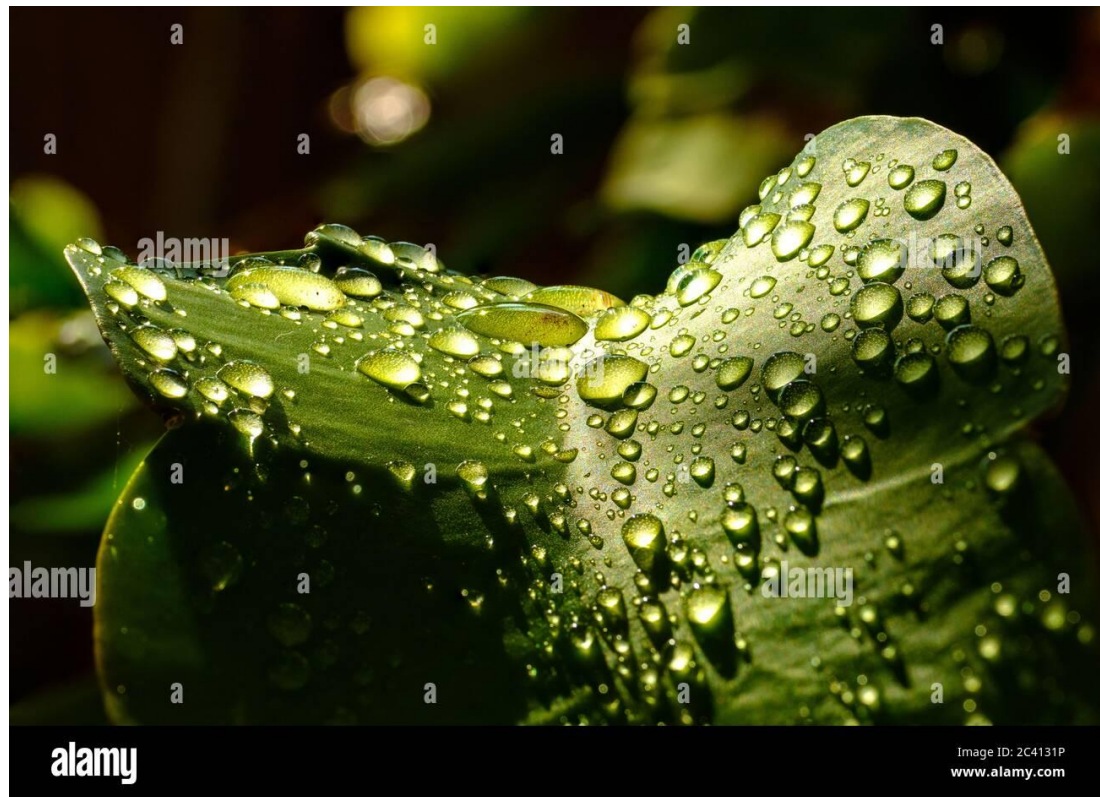
B. Simple Lipids (Fatty Acyl Esters)

2. Waxes:

- **Structure:** A long-chain fatty acid linked by an ester bond to a long-chain alcohol.
- **Function:** Waxes are extremely water-repellent and provide protective coatings for plants (on leaves), animals (on skin and fur), and insects (in exoskeletons).

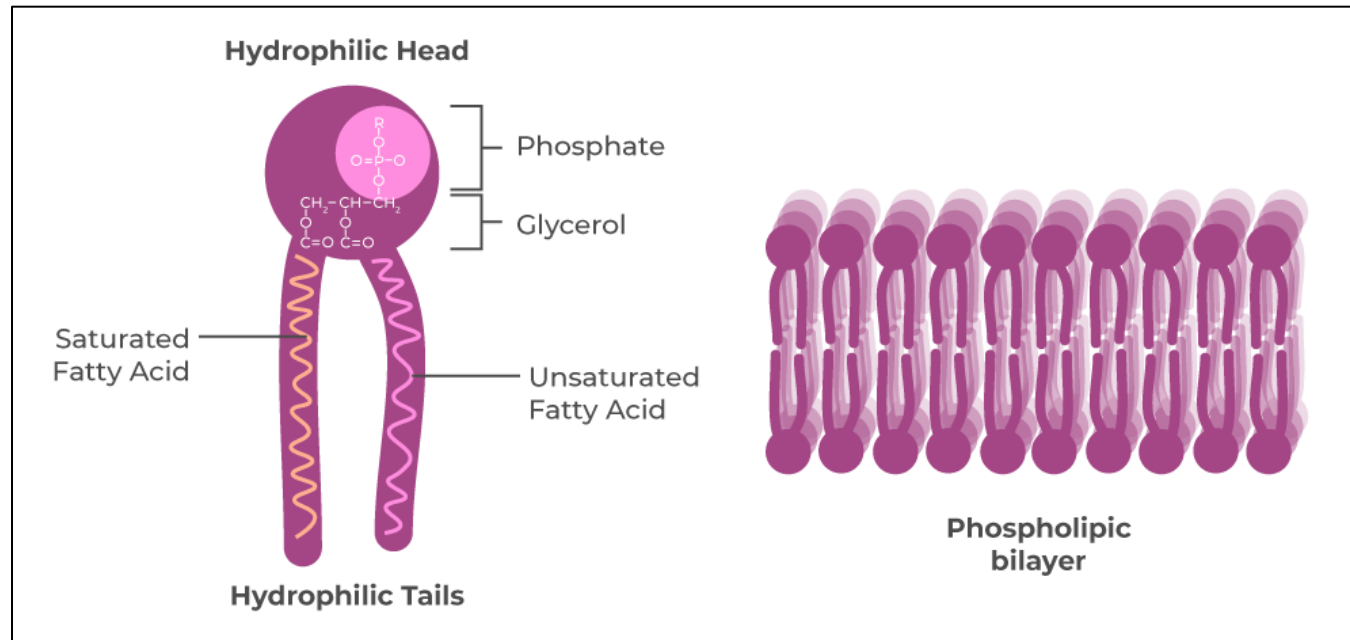
Example: Beeswax.





C. Complex Lipids (Membrane Lipids)

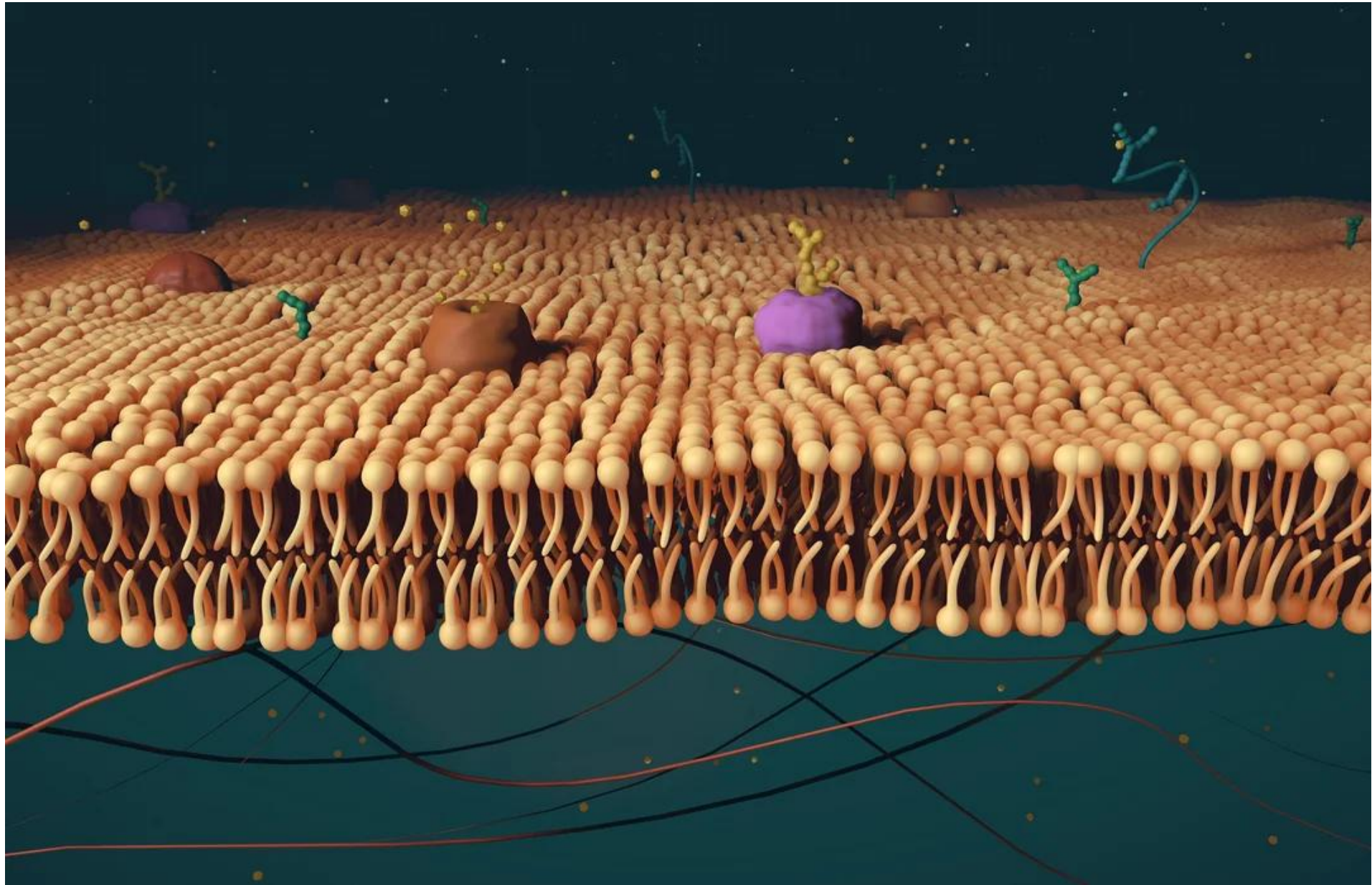
- These lipids are the major structural components of biological membranes.
- They are **amphipathic**, meaning they have both a hydrophilic (water-loving) and a hydrophobic (water-fearing) region.



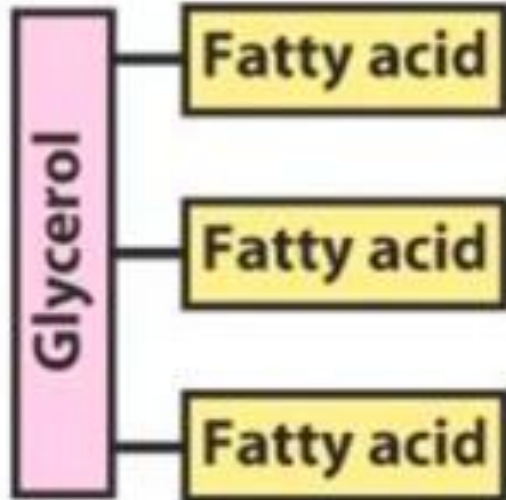
C. Complex Lipids (Membrane Lipids)

1. Phospholipids: The most abundant membrane lipids.

- **General Structure:** An amphipathic molecule with a **hydrophilic "head"** (containing a phosphate group) and two **hydrophobic "tails"** (fatty acid chains).
- **Sub-classes of Phospholipids:**
 - **Glycerophospholipids:** Have a **glycerol** backbone. Two fatty acids are esterified to C1 and C2, and a phosphate group is attached to C3. The phosphate group is often linked to a polar head group (e.g., choline, serine, ethanolamine), defining the specific type of phospholipid (e.g., phosphatidylcholine).
 - **Sphingolipids:** Have a **sphingosine** backbone instead of glycerol. One fatty acid is attached to the sphingosine, and the polar head group is attached to the C1 carbon.

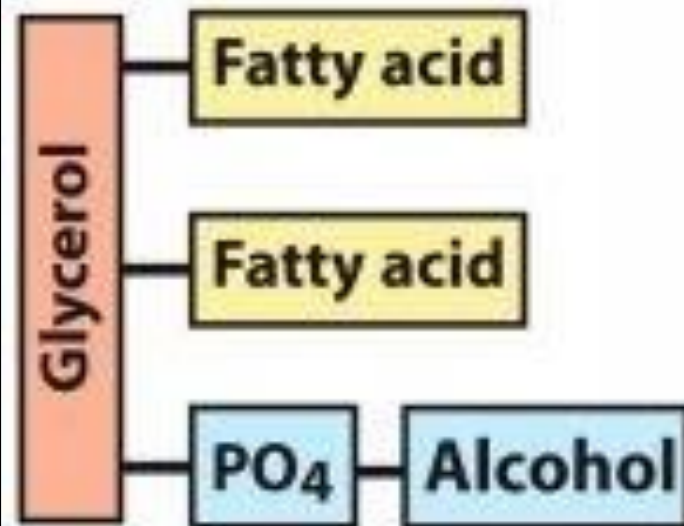


Triacylglycerols

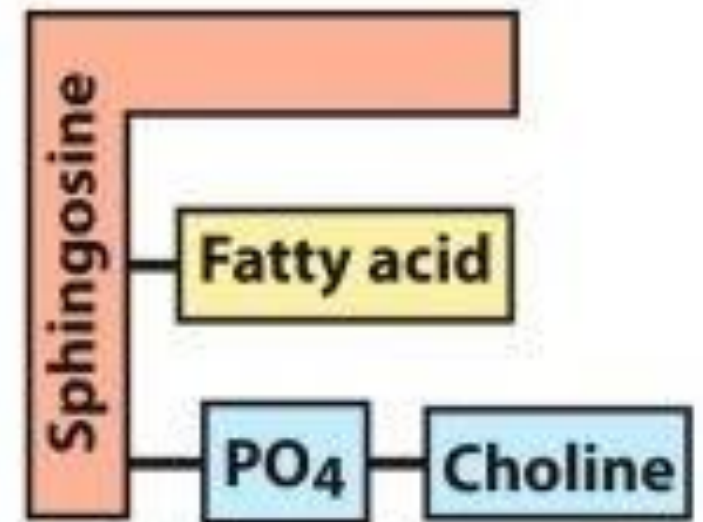


Phospholipids

Glycerophospholipids



Sphingolipids



C. Complex Lipids (Membrane Lipids)

2. Glycolipids:

- **Structure:** Similar to sphingolipids, with a **sphingosine** backbone and a single fatty acid tail. However, their hydrophilic head group consists of one or more **sugar residues** (monosaccharides or oligosaccharides) instead of a phosphate group.
- **Function:** Found on the outer surface of cell membranes, where they play a crucial role in cell recognition, cell-cell interactions, and blood typing.
- **Examples:** Cerebrosides (one sugar residue), Gangliosides (complex oligosaccharide chains).

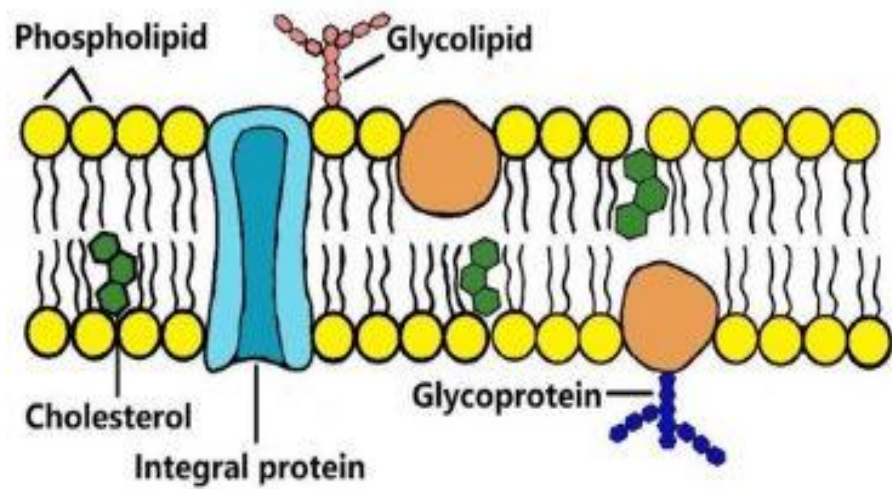
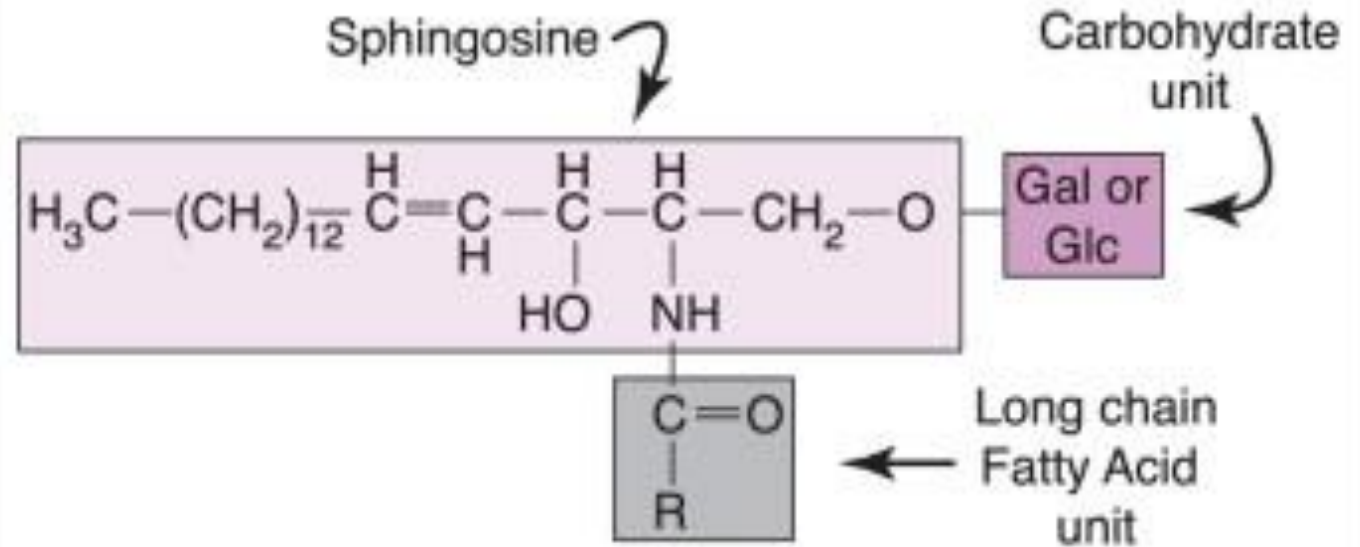


Fig: Structure of glycolipid

A Cerebroside (Glycolipid)



D. Steroids

Steroids are a class of lipids that are structurally distinct from fatty acid-based lipids.

- **Structure:** Characterized by a core structure of **four fused rings** (three six-membered and one five-membered ring).
- **Function:** Diverse, acting as hormones and structural components.

D. Steroids

Cholesterol

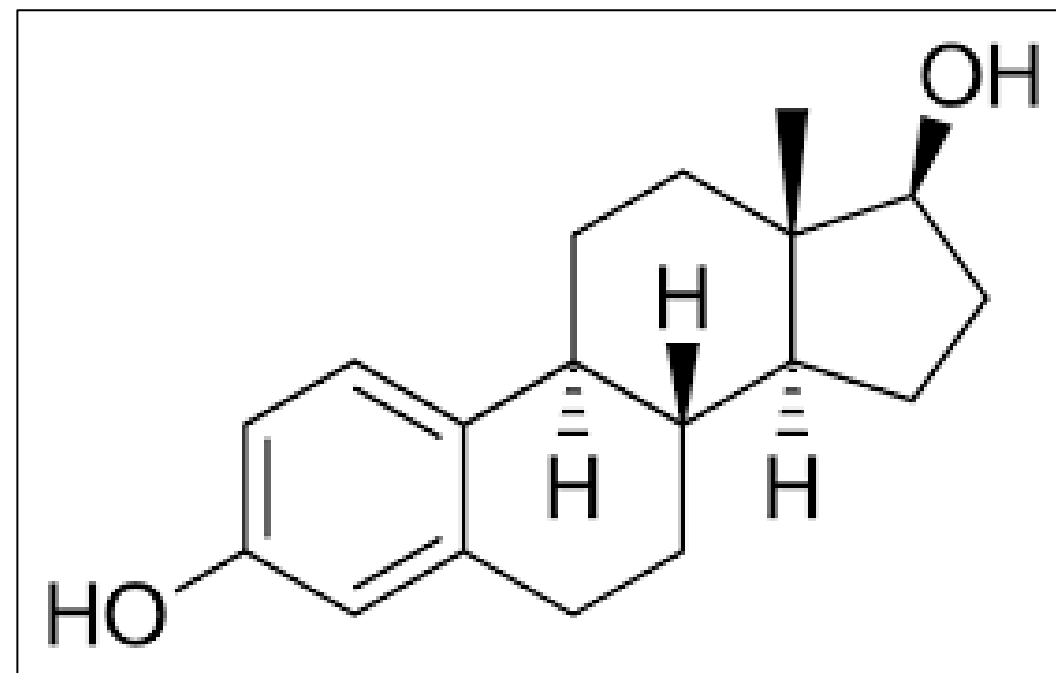
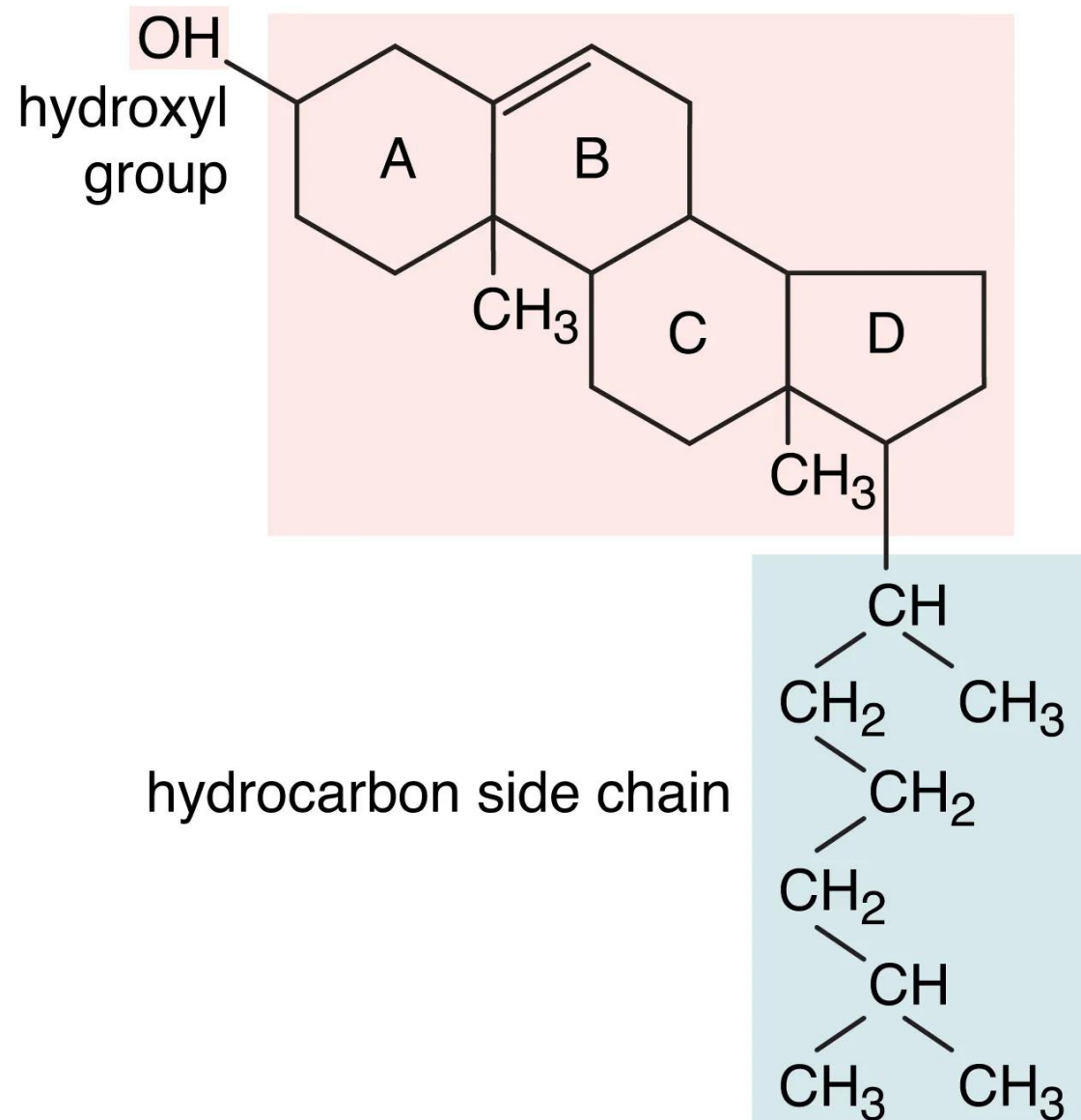
- **Structure:** An amphipathic molecule with the four-ring structure, a hydrocarbon tail, and a hydroxyl group on one of the rings.
- **Function:** A crucial component of animal cell membranes. It acts as a fluidity buffer, preventing the membrane from becoming too rigid at low temperatures or too fluid at high temperatures. It is also the precursor for all other steroid hormones.

Steroid Hormones

- **Structure:** Synthesized from cholesterol by modifying the side chains on the four-ring structure.
- **Function:** Act as potent signaling molecules (e.g., testosterone, estrogen, cortisol, aldosterone).

cholesterol

steroid nucleus



E. Other Important Lipid Classes

- **Eicosanoids:** Derived from a 20-carbon fatty acid called **arachidonic acid**. These are powerful, short-range signaling molecules that include prostaglandins, thromboxanes, and leukotrienes. They are involved in inflammation, blood clotting, and smooth muscle contraction.
- **Fat-Soluble Vitamins (A, D, E, K):** These are essential nutrients that cannot be synthesized by the body and are transported and stored in lipids.

General Properties of Lipids

Amphipathic Nature

- The presence of both polar and nonpolar regions in molecules like phospholipids and cholesterol is the fundamental reason they can form biological membranes.
- When placed in water, these molecules spontaneously arrange themselves into a **lipid bilayer**, with their hydrophilic heads facing the aqueous environment and their hydrophobic tails tucked away in the nonpolar interior.

General Properties of Lipids

Melting Point: The melting point of fatty acids and triacylglycerols is directly influenced by their structure.

- **Chain Length:** Longer chains have higher melting points due to increased Van der Waals interactions.
- **Degree of Saturation:** Saturated fatty acids have higher melting points than unsaturated fatty acids of the same length because their straight chains can pack more closely together. The kinks in *cis*-unsaturated chains prevent tight packing, requiring less energy to melt.

General Properties of Lipids

Chemical Reactivity:

- **Esterification:** The reaction that forms ester bonds (e.g., between fatty acids and glycerol in triacylglycerols and phospholipids).
- **Hydrolysis:** The reverse reaction, breaking ester bonds with the addition of water (e.g., during fat digestion by lipases).
- **Hydrogenation:** The process of adding hydrogen to unsaturated fatty acids to reduce the number of double bonds, making them more saturated and solid. This can inadvertently produce *trans* fats.
- **Oxidation (Rancidity):** The double bonds in unsaturated fatty acids are susceptible to oxidation, which can generate volatile aldehydes and ketones, leading to an unpleasant odor and taste (rancidity).

THANK YOU